



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

12 September 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 9
2	/ 9
3	/ 11
4	/ 10
5	/ 11
6	/ 9
7	/ 10
8	/ 8
9	/ 12
10	/ 11
Total	/ 100

QUESTION 1

Fig. 1.1 is an electron micrograph of part of a eukaryotic cell.



Fig. 1.1

(a) Identify the structures **J** and **K**. [2]

J

K

(b) Describe two structural features shown in Fig. 1.1 that identify **G** as the Golgi apparatus and **not** the rough endoplasmic reticulum. [2]

1.

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2.

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The Golgi apparatus is an organelle that is important for many functions. One example is the formation of lysosomes.

Before lysosomal hydrolases can function in the lysosome, they must be sorted from other proteins. Fig. 1.2 shows how a glycosylated lysosomal hydrolase is sorted at the trans face of the Golgi apparatus.

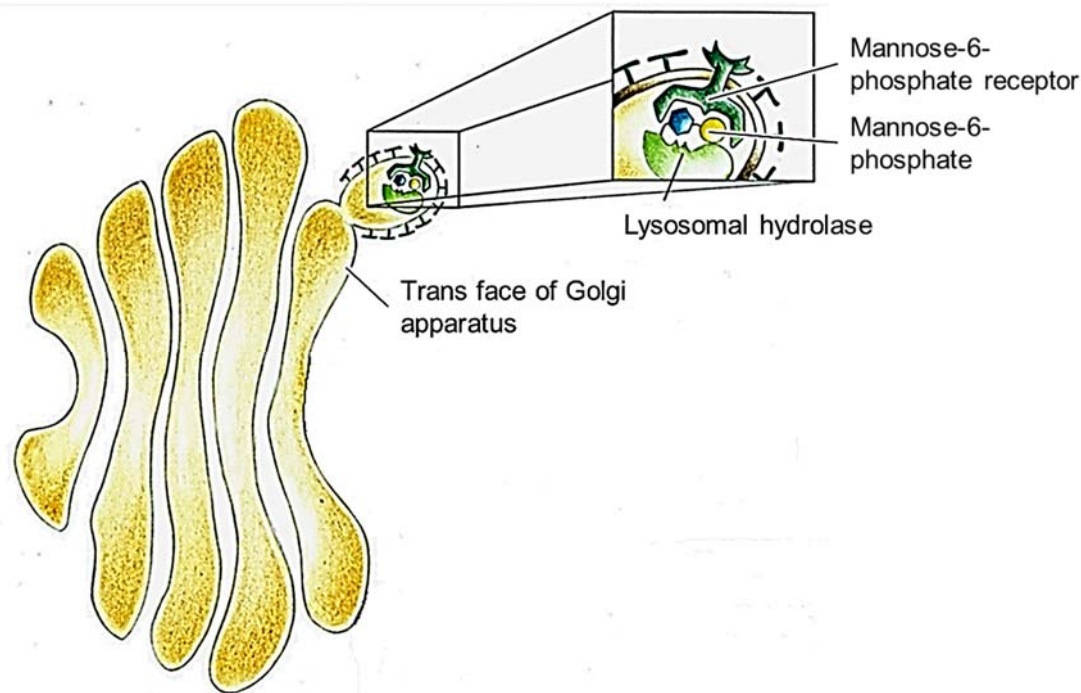


Fig. 1.2

(c) Outline the pathway taken by the lysosomal hydrolase from its site of synthesis to the trans face of the Golgi apparatus. [3]

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(d) With reference to Fig. 1.2, suggest how the Golgi apparatus is able to specifically sort the lysosomal hydrolase from other proteins. [2]

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QUESTION 2

(a) Contrast the processes of facilitated diffusion and active transport.

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(b) A group of students investigated the uptake of chloride ions in barley plants. They divided the plants into two groups and placed their roots in solutions containing radioactive chloride ions.

- Group A plants had a substance that inhibited respiration added to the solution.
- Group B did not have the substance added to the solution.

The students calculated the total amount of chloride ions absorbed by the plants every 15 minutes. Their results are shown in Fig. 2.1.

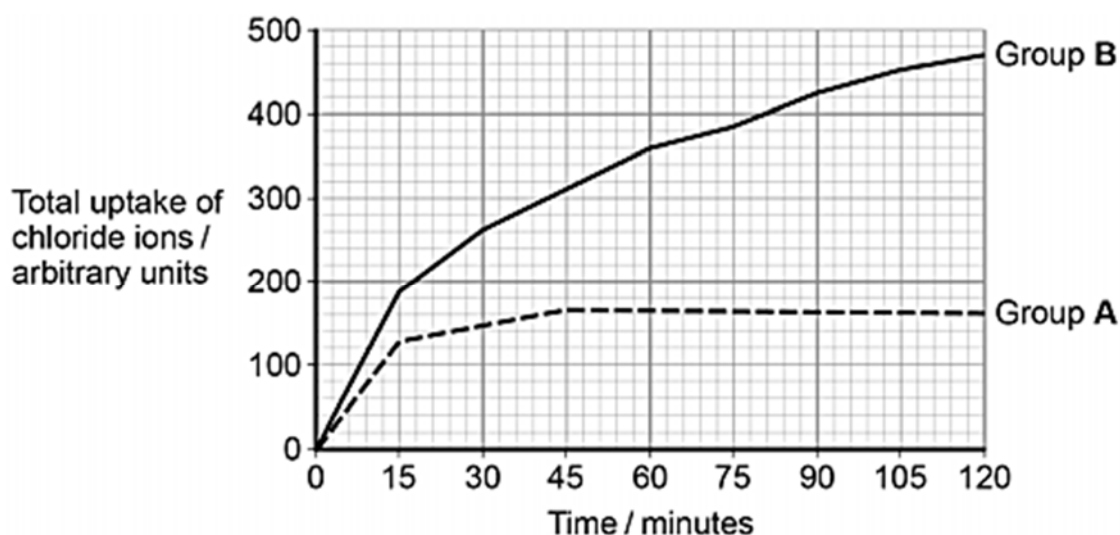


Fig. 2.1

(i) Calculate the ratio of the **rate** of uptake of chloride ions in the first hour to the **rate** of uptake of chloride ions in the second hour for group B plants. [2]

Ratio:

(ii) Explain the results shown in Fig. 2.1.

[4]

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[Total: 9]

QUESTION 3

Bacteria synthesize the amino acid isoleucine from threonine in a series of five enzyme-catalysed steps. When the concentration of isoleucine increases in the cell, isoleucine inhibits threonine deaminase, the first enzyme in the pathway.

Fig. 3.1 shows these five steps in the pathway to synthesise isoleucine.

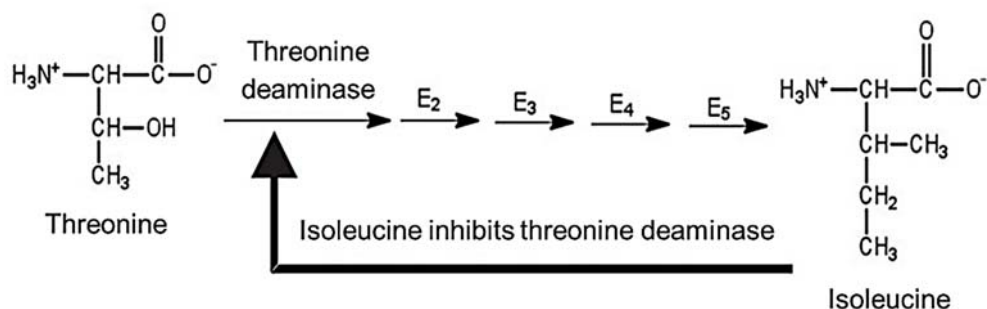


Fig. 3.1

- (a) Explain how and why isoleucine inhibits threonine deaminase when the concentration of isoleucine increases. [4]

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These amino acids are used to synthesise the peptidoglycan cell walls of bacteria. Peptidoglycans are long chains of the sugars N-acetylmuramic acid (NAM) and N-acetylglucosamine (NAG) which alternate along the chain. A short peptide chain is attached to each NAM and these form cross-links with similar peptide chains from adjacent strands.

Fig. 3.2 shows a diagram representing part of a peptidoglycan structure.

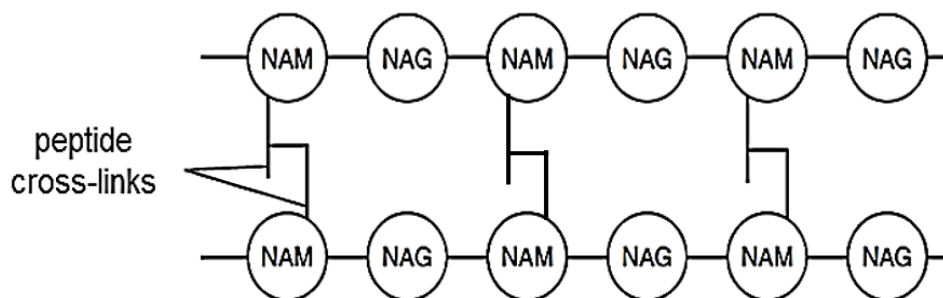


Fig. 3.2

(b) State two differences between the structure of the peptidoglycan cell wall and that of the plant cell wall. [2]

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The formation of these peptidoglycan cross-links are catalysed by the enzyme transpeptidase. Penicillin inhibits the action of this enzyme.

(c) Describe the mode of action of penicillin on bacteria. [3]

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Bacteria may be classified as Gram-positive or Gram-negative bacteria.

Fig. 3.3 shows a diagram of part of the cell walls of both Gram-positive and Gram-negative bacteria.

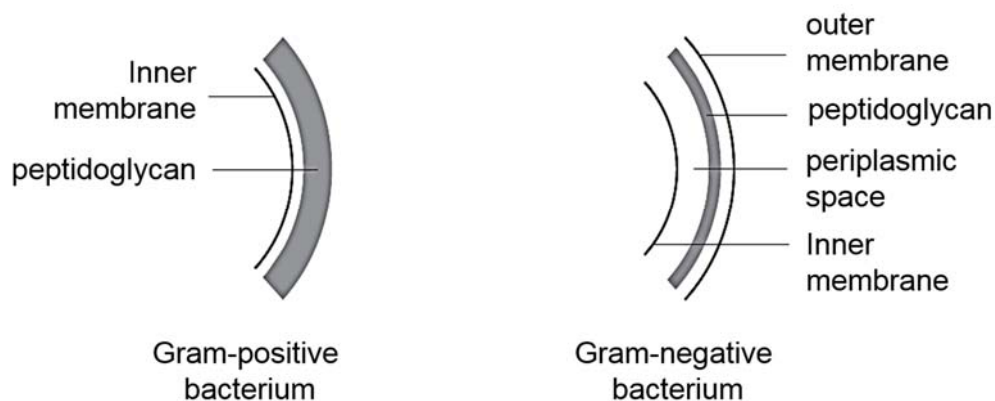


Fig. 3.3

(d) Predict the effectiveness of penicillin on killing the two types of bacteria. Explain your answer. [2]

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[Total: 11]

QUESTION 4

Palisade cells have both chloroplasts and mitochondria. Exchanges between a mitochondrion, a chloroplast and the cytoplasm surrounding them are shown in Fig 4.1.

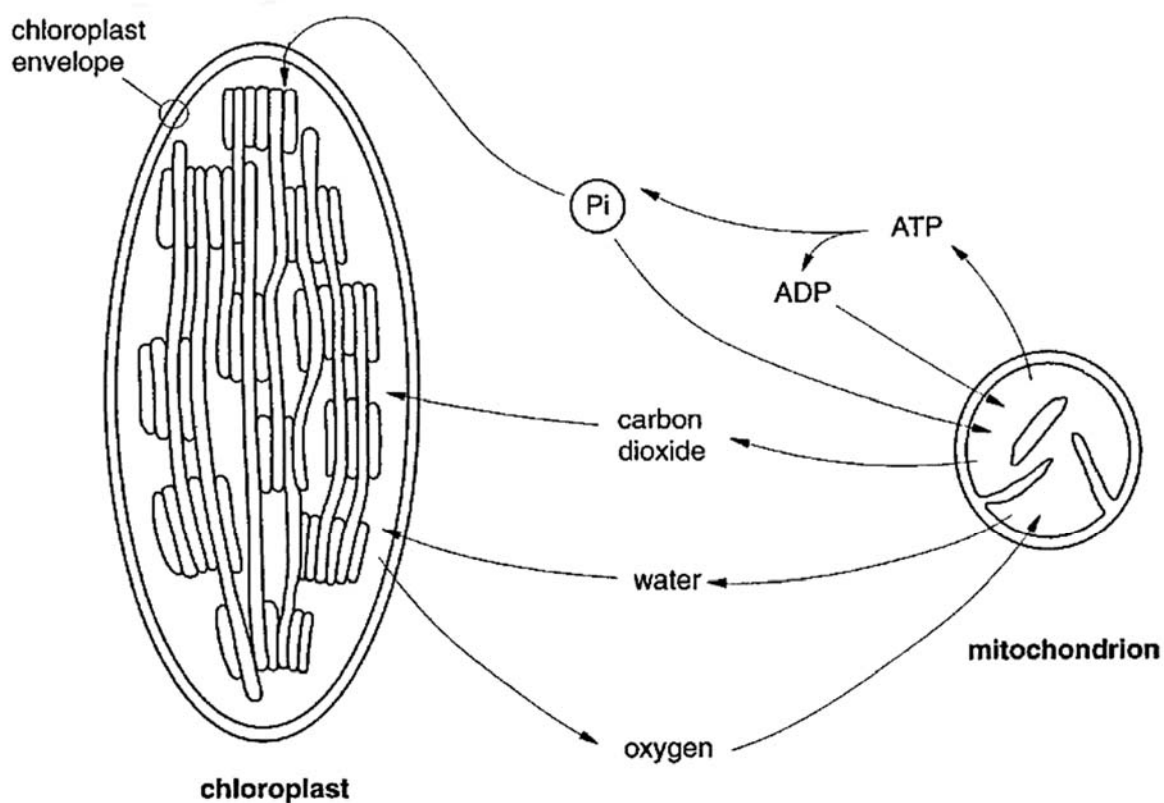


Fig. 4.1

(a) Explain why oxygen is needed for the production of ATP in the mitochondrion.

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A leafy shoot is sealed inside a transparent container. The concentration of oxygen in the atmosphere within this container is measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(b) Using Fig. 4.1, explain how it is possible for the oxygen concentration to remain constant. [2]

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(c) (i) Sketch using the axes below, a graph showing the relationship between light intensity and the change in atmospheric oxygen concentration in the container with the leafy shoot. [1]

(ii) Indicate with an 'X' the light intensity where the oxygen concentration remains constant. [1]



(d) Both the chloroplast and mitochondrion synthesise ATP.

Describe the fates of ATP synthesised by these two organelles. [2]

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[Total: 10]

QUESTION 5

Fig. 5.1 shows the stages involved in the synthesis of a p53 protein in a eukaryotic cell.

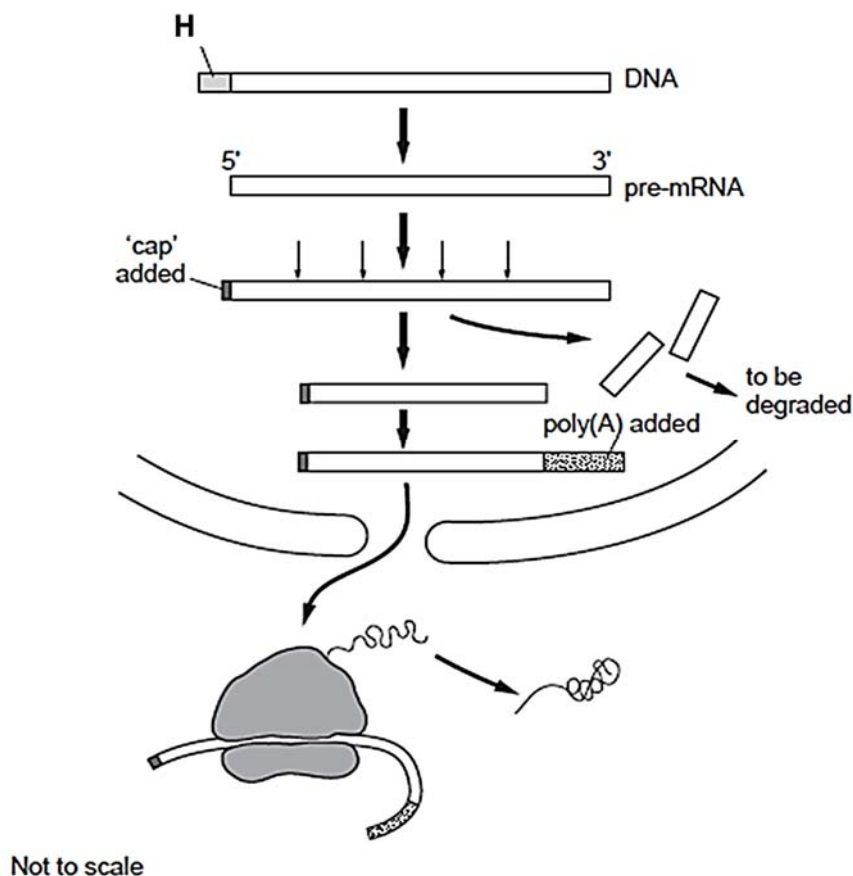


Fig. 5.1

(a) Name the structure H.

[1]

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(b) Outline the events that occur in the nucleus of the cell to produce a functional molecule of mRNA encoding a p53 protein.

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(c) In a particular defective cell, all the pre-mRNA encoding the p53 protein were **not** cut.

(i) Explain how this would affect the function of the p53 protein. [4]

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(ii) Explain why this may not lead to cancer. [2]

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[Total: 11]

End of booklet 1
Proceed to booklet 2

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Candidate Name

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Civics Group Index Number

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Meridian Junior College
JC2 Preliminary Examination 2018
H2/9744 Biology

Booklet 2
Questions 6 to 10

Answer **all** the questions in this section.

For Examiner's Use	
6	/ 9
7	/ 10
8	/ 8
9	/ 12
10	/ 11

QUESTION 6

STAT proteins are cytoplasmic transcription factors that play important roles in the development and differentiation of many cell types. Upon external stimulation, STAT protein is activated from its inactive form and binds to another activated STAT protein to form a dimer. This protein dimer then translocates to the nucleus and regulates the expression of other genes as shown in Fig. 6.1.

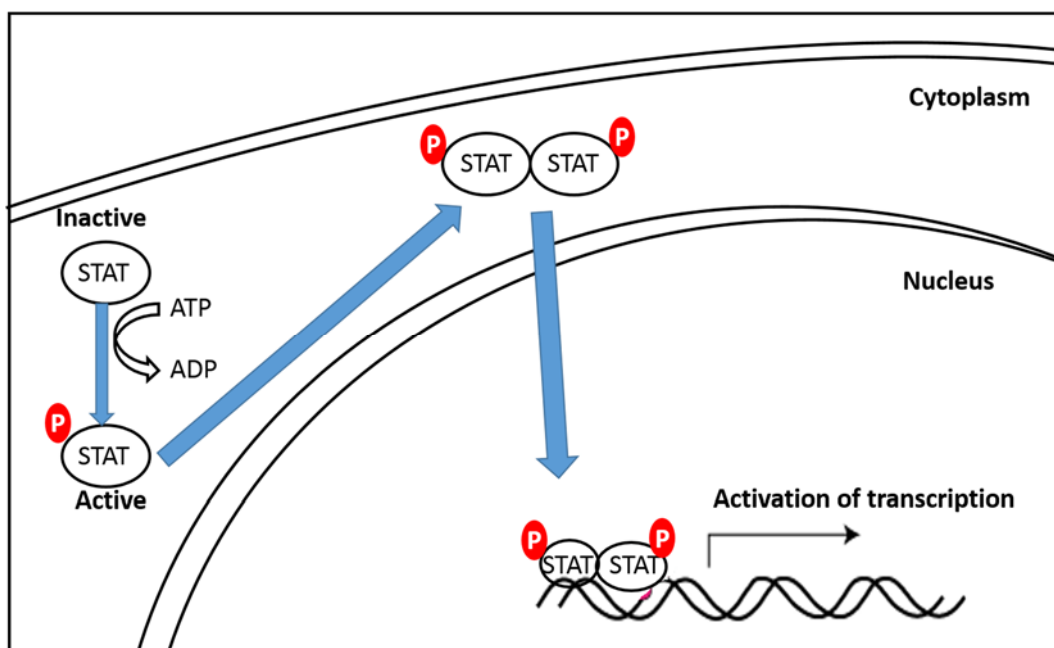


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) explain how the inactive STAT protein is converted to its active form.

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(ii) besides converting the inactive form of STAT protein to its active form, describe how the level of the inactive STAT protein may be controlled after its production.

[2]

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STAT5 gene, a member of the *STAT* family, is widely expressed in hematopoietic stem cells (HSC) to regulate the self-renewal and differentiation of the stem cells.

Fig. 6.2 shows the differentiation of HSC leading to the formation of different cell types such as T cell and B cell.

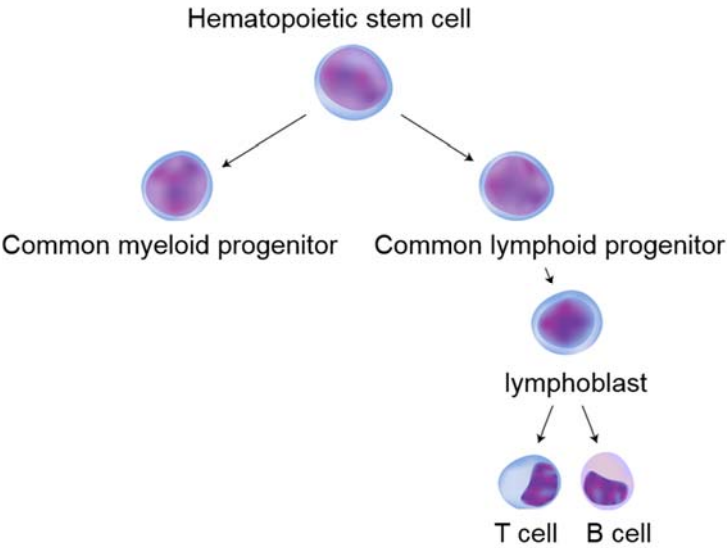


Fig. 6.2

(b) Explain how the different cell types such as T cell and B cell can arise from a single hematopoietic stem cell. [3]

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(c) In humans there are different forms of *STAT5* protein, each plays a slightly different role in different cell types.

Explain how the same *STAT5* gene can produce different forms of *STAT5* protein. [2]

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[Total: 9]

QUESTION 7

Apart from animal cells, bacteria such as *E. coli* can also be infected by viruses. An example of such viruses is the T4 bacteriophage.

Fig. 7.1 shows a T4 bacteriophage which has been modified to contain the DNA of Mu bacteriophage, a temperate phage. The DNA of Mu bacteriophage is able to integrate into the bacterial chromosome.

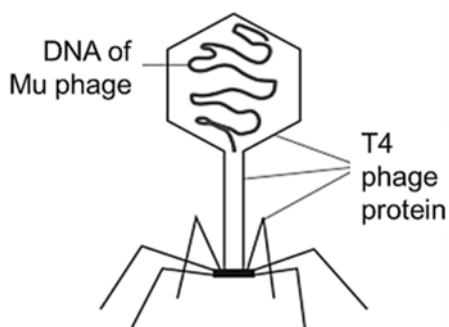


Fig. 7.1

(a) (i) Describe how DNA of Mu bacteriophage enters an *E. coli* host cell. [2]

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(ii) Explain the importance of the phage genome in integrating the phage DNA into the host chromosome. [2]

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(b) Describe how the reproduction of these modified T4 bacteriophages could contribute to genetic variation in a colony of *E. coli*. [4]

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(c) Some scientists believe that bacteriophages could be used to treat people with bacterial infection.

Suggest the properties of the bacteriophages that would make this possible. [2]

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[Total: 10]

QUESTION 8

A researcher created an artificial plasmid that places the promoter (**P_{trp}**) and operator (**O_{trp}**) of the *trp* operon upstream of the *lac* structural genes and downstream of the CRP binding site of the *lac* operon, as shown in Fig. 8.1. The *trp R* gene, which codes for the *trp* repressor, is also inserted into the plasmid.

The plasmids were introduced into bacterial cells.

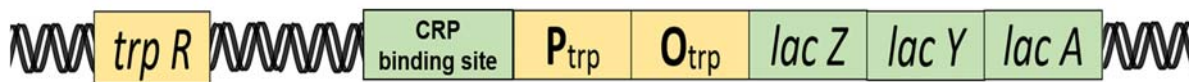


Fig. 8.1

- (a)** State whether the above operon is an inducible or a repressible system. [1]

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- (b)** Explain the condition needed for the expression of *lac ZYA* genes. [4]

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- (c)** With reference to Fig. 8.1, describe how this operon is positively regulated. [3]

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[Total: 8]

QUESTION 9

(a) Explain the meaning of the terms

[2]

(i) *locus*,

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(ii) *heterozygous*.

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Immunodeficiency disorders are caused when the immune system's ability to fight infectious disease is compromised. Bruton's syndrome is a rare immunodeficiency disorder that is characterized by a mutation in the Bruton's tyrosine kinase gene, which blocks the development of B cell. Individuals with Bruton's syndrome are vulnerable to repeated, potentially fatal, bacterial infections. To better understand the inheritance pattern of Bruton's syndrome, a family was followed for three generations as shown in Fig. 9.1.

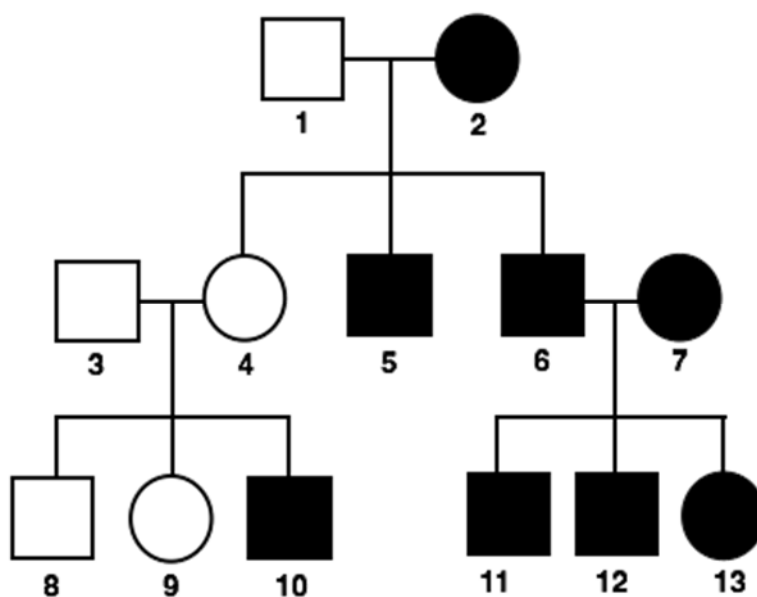


Fig. 9.1

(b) With reference to Fig. 9.1, predict and explain the most likely mode of inheritance of Bruton's syndrome. [3]

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(c) Individual 8 married a normal female whose father is affected.

(i) Using suitable symbols, construct a genetic diagram of the cross. [5]

(ii) State the probability that their first 2 children are all unaffected females. Show your working. [1]

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(d) Children with Bruton's syndrome have no or very low level of immunoglobulins in their blood. As such, they are vulnerable to repeated infections.

Suggest a possible long-term treatment for children with Bruton's syndrome. [1]

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[Total: 12]

QUESTION 10

Reproduction in seahorses, *Hippocampus*, is unusual as it is the male rather than the female that becomes pregnant. The male has a brood pouch located on its tail. The larger the male the larger the pouch. The female transfers unfertilised eggs into the pouch. The larger the female the more eggs are produced that can be transferred to the brood pouch. The male releases sperm onto the eggs and they are fertilised. The male carries the developing brood for a period of several weeks until he finally gives birth.

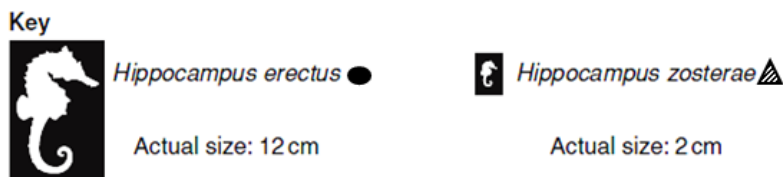
Research into seahorse population has revealed the following:

- They are monogamous. A male and female remain together for the whole mating season.
- Within a population, mates are selected by size. Large females mate with large males and small females mate with small males.
- Few intermediate sized individuals are produced and they have a low survival rate.

Two different species of seahorse are found in the coastal regions shown in Fig. 10.1. The ranges of these two seahorse species overlap in many areas of these waters.



Fig. 10.1



The two seahorse silhouettes are not drawn to scale.

(a) With reference to Fig. 10.1, explain how speciation has occurred in the seahorses. [5]

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(b) There are 4 different species of seahorses from the Atlantic Ocean. Samples of these species were collected and mitochondrial DNA was used to study the evolutionary relationship between the four species.

Explain why mitochondrial DNA is suitable for constructing phylogenetic relationships. [3]

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Seahorses and spider monkeys have evolved independently to possess a prehensile tail, allowing them to grasp and hold onto structures. Fig. 10.2 shows the prehensile tail structures in both seahorses and spider monkeys.



Fig. 10.2

- (c) A student concluded that the prehensile tail shared by both seahorses and spider monkeys is an example of homologous structures. Explain whether the conclusion made is valid. [3]

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[Total: 11]

End of Paper